From Latent-Collapse Diagnosis to a Stronger Baseline

L1-only particle-level embedding model for anomaly detection

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the NextGen WP3 OpenLab -

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From diagnosing collapse to selecting a stronger baseline

- **Last week** showed the latent collapse is stable across seeds and not resolved by architecture changes alone.
- **This round asks**: can we select a stronger supervised configuration, and does more physics structure change the picture?
- **Three controlled axes**: best configuration, label granularity, and loss-objective design



Selected for balance, not for maximizing one metric

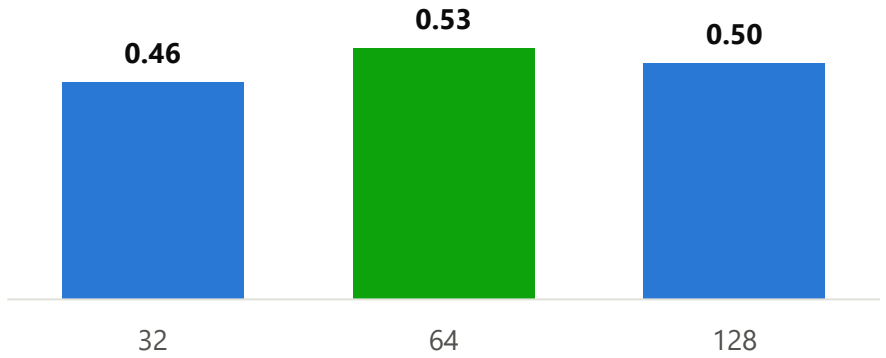
BASELINE CONFIGURATION

- Updated Collide-2V
- L1-only inputs
- top_pt selection
- MHA model
- latent_dim=64
- ff_dim=128
- num_heads=8
- 2 attention blocks
- pooling=mean
- classifier=linear
- var_reg_weight=0.01
- early-stop patience 10

var_reg=0.01 selection (5-class benchmark):

SETUP	ACCURACY	HH4B F1	EFF. RANK	EVAL_Q
CE only	0.619	0.713	4.03	0.180
+ var_reg 0.01	**0.624**	**0.717**	**6.55**	**0.196**
+ var_reg 0.03	0.613	0.711	5.46	0.189

LATENT_DIM=64 SELECTION (GRANULAR-LABEL SWEEP)



DY F1 vs. latent_dim

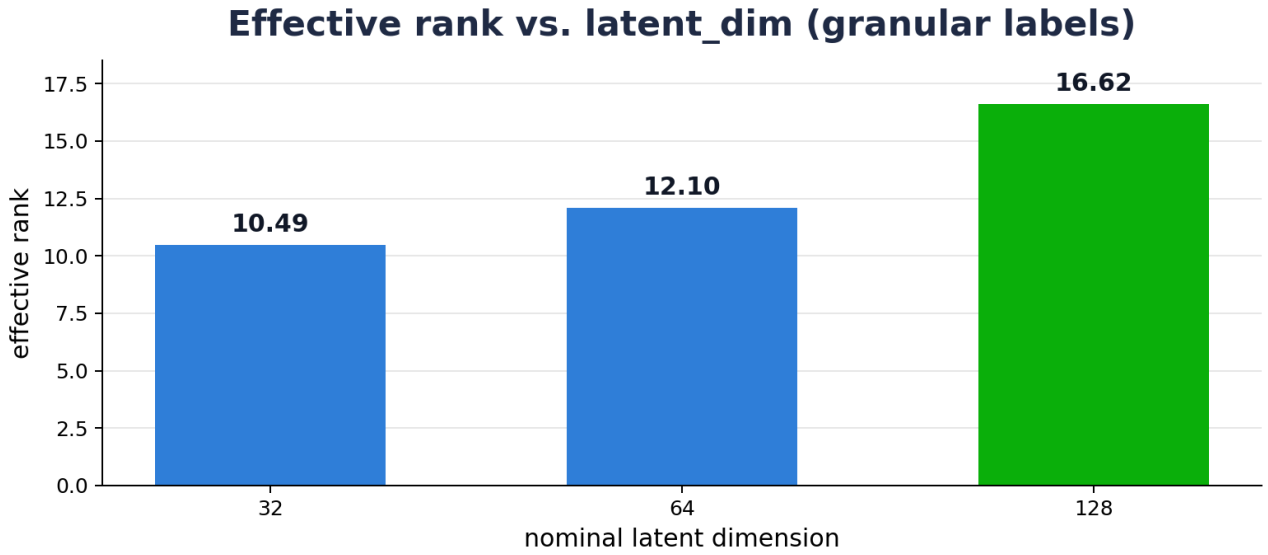
LATENT_DIM	HH4B F1	EVAL_Q
32	0.556	0.170
64	**0.561**	**0.171**
128	0.550	0.0008

64 avoids weaker DY (32) and the near-zero eval-quality run (128).

Finer final-state labels activate more latent structure

Split Top -> tt_hadronic / tt_leptonic / tt_semileptonic · WW -> WW_hadronic / WW_leptonic / WW_semileptonic (10 classes total)

- **Classification became harder** than the 5-class setup: accuracy dropped to roughly 0.46-0.47 (from ~0.62) — expected, since final-state splits are less separable.
- **Effective latent dimensionality rose** well beyond the 5-class case: ~10.5 at latent_dim=32, ~12.1 at latent_dim=64, ~16.6 at latent_dim=128
- **Interpretation:** more physically distinct target classes appear to activate more of the latent space — this supports the hypothesis that part of the earlier collapse reflected too few label directions.



Granular labels activate more latent structure: effective rank rises from ~10.5 to ~16.6 as nominal latent capacity increases.

A broader, more realistic SM benchmark for HH4B

PREVIOUS 5-CLASS BENCHMARK

- MinBias
- Top
- QCD
- DY
- HH4B (anomaly)

NEW 10-CLASS SM SUPERCLASS BENCHMARK

- MinBias
- QCD
- DY
- Top
- W+jets
- WW
- WZ
- ttW
- ttZ
- HH4B (anomaly)

Five new SM processes added (green) — W+jets, WW, WZ, ttW, ttZ.

- **HH4B is evaluated** against this larger, more diverse SM background, not a small 4-process set.
- Same baseline configuration as Slide 3 (latent_dim=64, ff_dim=128, heads=8, 2 blocks, mean pooling, linear head, var_reg=0.01), 100k events/class.
- **Message**: a harder classification problem, but a more realistic proxy for how an embedding would need to behave against real SM backgrounds.

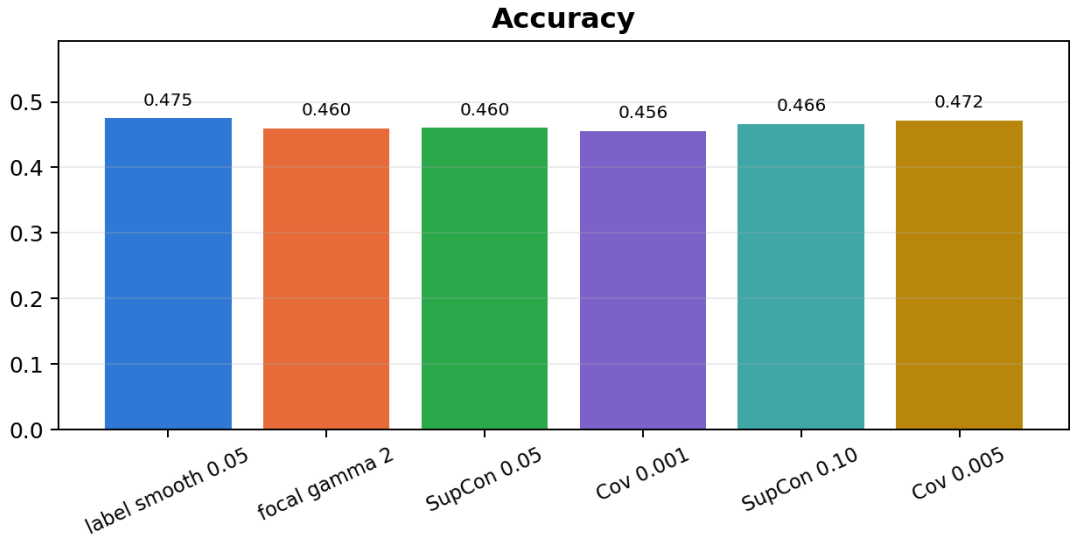
Testing six CrossEntropy-based objective variants

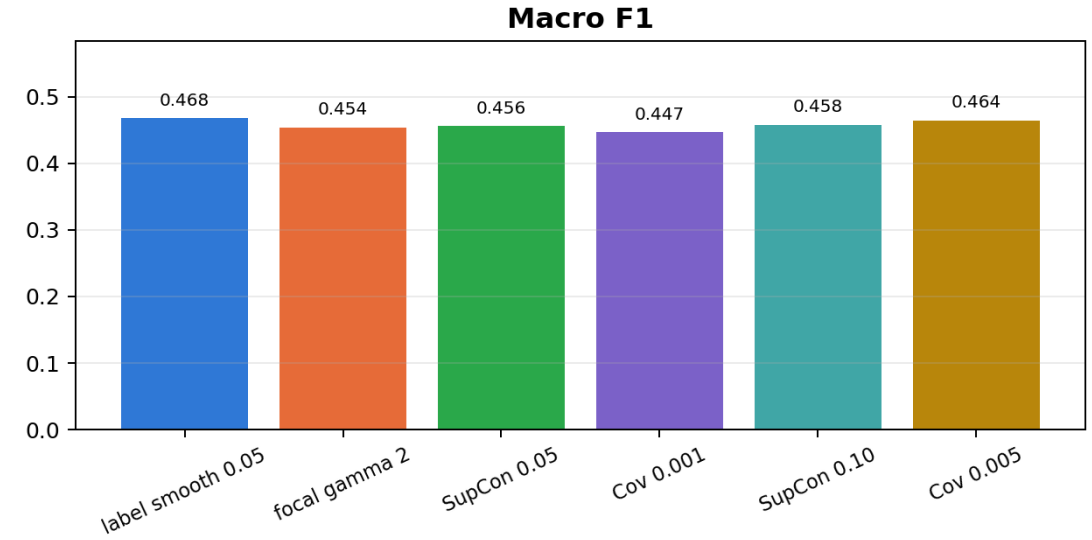
All variants: broader 10-class superclass setup · latent_dim=64 · 100k events/class · same architecture as Slide 3

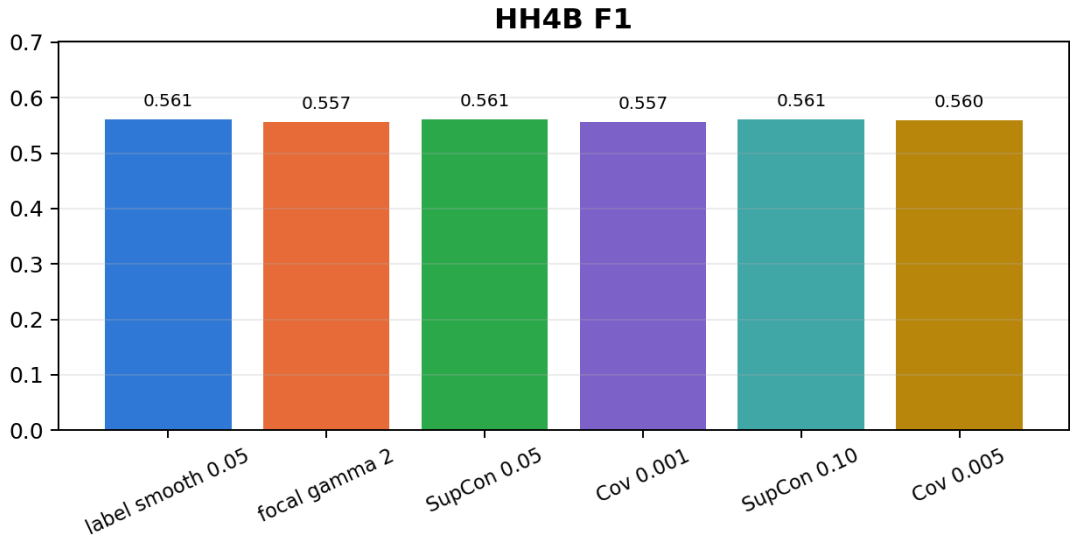
#	VARIANT	DESCRIPTION
1	CE + variance reg.	reference point, carried from Week 2
2	CE + label smoothing (0.05) + variance	softens hard class targets
3	Focal loss (g=2) + variance	down-weights easy examples
4	CE + SupCon (0.05) + variance	supervised contrastive term, weight 0.05
5	CE + variance + covariance reg. (0.001)	decorrelates latent dimensions
6	Stronger SupCon (0.10) / covariance (0.005)	stronger versions of variants 4 & 5

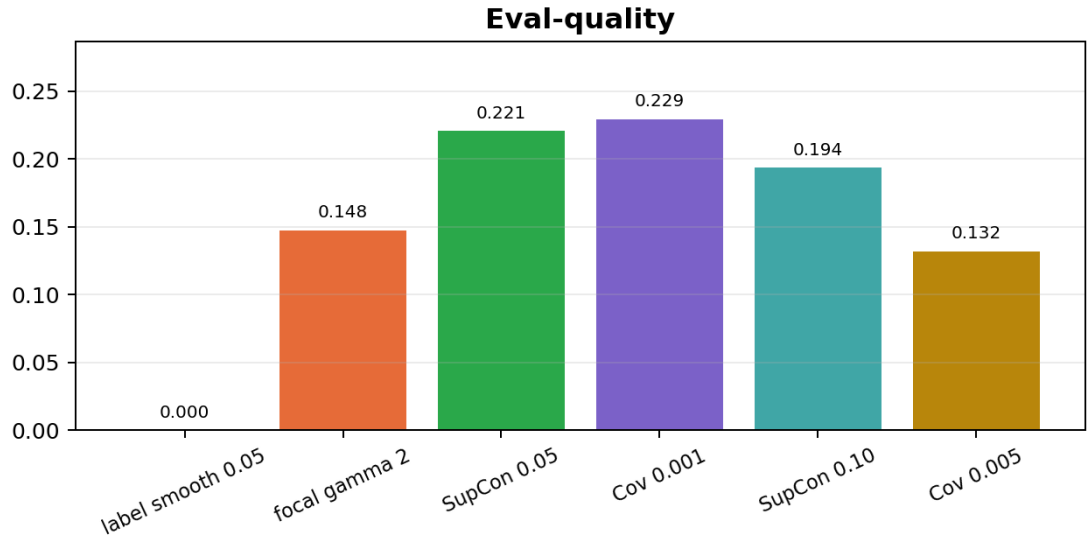
- **Goal**: test whether changing the training objective can activate more of the latent space without sacrificing classification or HH4B performance.
- A controlled comparison at fixed architecture and label set — only the loss function changes across runs.

Superclass loss-objective comparison



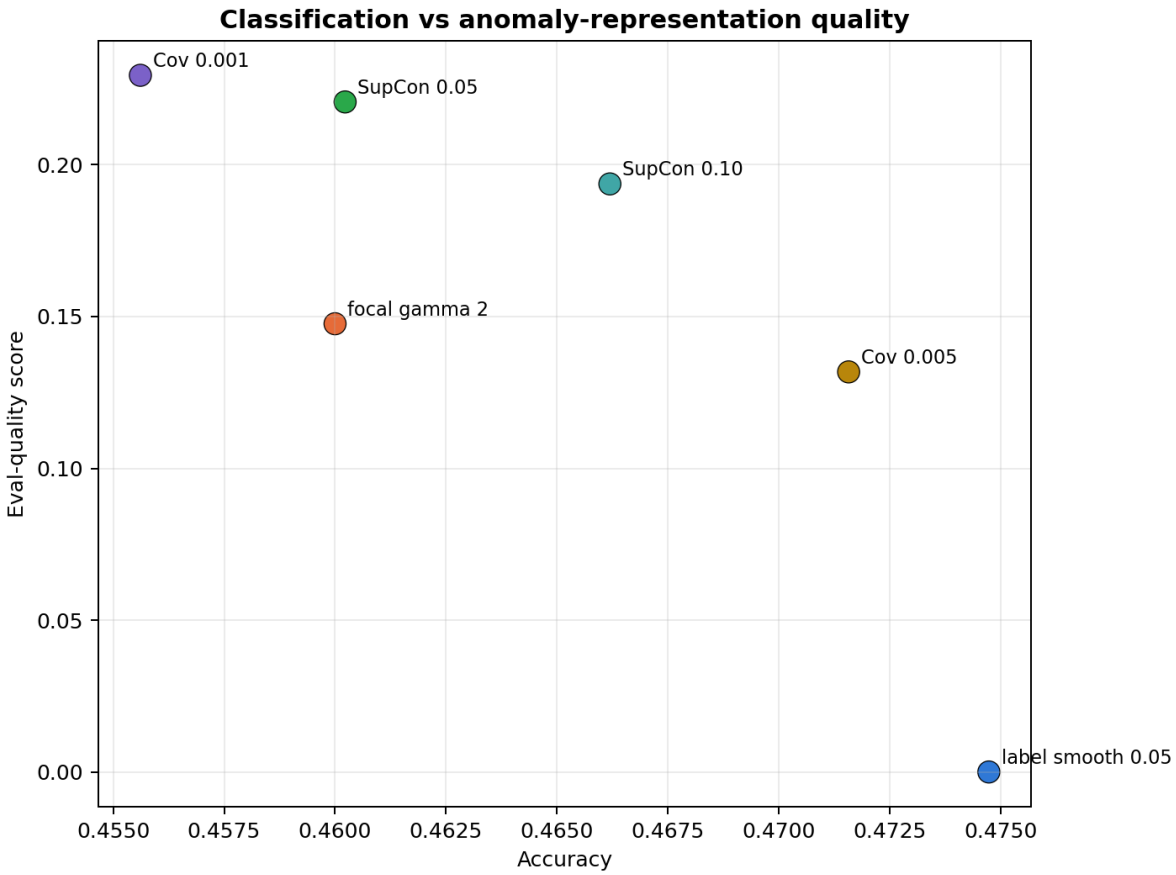






Testing six CrossEntropy-based objective variants

- Covariance-regularized variants give by far the largest effective rank (~29.1 at cov=0.001, ~27.5 at cov=0.005) and highest eval-quality (0.229 at cov=0.001) — but not the best accuracy or HH4B F1.
- SupCon variants give the best DY F1 (0.519 at SupCon=0.05) with only modest effective rank (~2.8-3.6) — nearly the opposite pattern.
- **Label smoothing** gives the highest accuracy (0.475) but eval-quality near zero (0.00004) — a clear task/geometry disagreement.
- HH4B F1 is essentially flat across all six variants (0.557-0.561) — no variant produced a clear anomaly-detection win.
- Metrics disagree with each other: accuracy, HH4B F1, eval-quality, and effective rank each capture a different property.



Best classification candidate: label smoothing + variance
Best anomaly-quality candidate: CE + variance + covariance, cov_weight=0.001
Best balanced candidate: CE + SupCon 0.05 + variance

Current interpretation

- **Latent collapse does not appear to be purely architectural** — Last week already ruled out simple capacity, depth, and width fixes.
- **CrossEntropy alone appears to encourage compression** toward a small number of class-separating directions — a working hypothesis, not a proven mechanism.
- **Granular, physically motivated labels activate more latent structure**, consistent with label granularity — not only latent_dim — shaping effective dimensionality.
- **Variance/covariance regularization can help**, but requires careful tuning and does not uniformly improve HH4B separation.
- **The broader SM superclass setup is harder but more realistic** as an anomaly-detection benchmark.